

Cloud Computing (Lab 12)

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BSE-V-B

Lab 12 — Terraform Provisioners, Modules & Nginx Reverse Proxy/Load Balancer

Objective

- Use GH CLI inside a Codespace
- Organize Terraform code into separate files
- Practice Terraform provisioners (user_data, remote-exec, file, local-exec)
- Create reusable Terraform modules for subnet and webserver
- Configure Nginx as reverse proxy and load balancer
- Implement SSL/TLS with self-signed certificates
- Configure Nginx caching
- Implement high availability with backup servers

Task 0 — Lab Setup (Codespace & GH CLI)

Screenshots Required:

- task0_codespace_create_and_list.png

```
C:\Users\malik\Lab12>gh codespace create --repo toobashafique065/Lab12 --machine basicLinux32gb
@ Codespaces usage for this repository is paid for by toobashafique065
congenial-space-winner-wrq7qxpr65g29vj4

C:\Users\malik\Lab12>gh codespace list
NAME                                DISPLAY NAME      REPOSITORY          BRANCH  STATE    CREATED AT
fuzzy-bassoon-4jp7p65jj46p...     fuzzy bassoon     toobashafique065/Lab9  main*   Shutdown about 5 days ago
potential-halibut-4jp7p65j...     potential halibut toobashafique065/Lab10 main*   Shutdown about 4 days ago
automatic-fortnight-pj5r59...     automatic fortnight toobashafique065/Lab11 main*   Shutdown about 14 hours ago
congenial-space-winner-wrq...     congenial space winner toobashafique065/Lab12 main    Available less than a minute ago
```

- task0_codespace_ssh_connected.png

```
C:\Users\malik\Lab12>gh codespace ssh -c congenial-space-winner-wrq7qxpr65g29vj4
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@toobashafique065 [ /workspaces/Lab12 (main) ] $ mkdir -p ~/Lab12
```

Task 1 — Organize Terraform Code

Screenshots Required:

- task1_project_directory.png

```
@toobashafique065 [ ] /workspaces/Lab12 (main) $ mkdir -p ~/Lab12
@toobashafique065 [ ] /workspaces/Lab12 (main) $ cd ~/Lab12
```

- task1_files_created.png

```
@toobashafique065 [ ] ~/Lab12 $ touch main.tf variables.tf outputs.tf locals.tf terraform.tfvars entry-script.sh
@toobashafique065 [ ] ~/Lab12 $ ls -la
total 12
drwxrwxr-x 2 codespace codespace 4096 Jan  2 09:53 .
drwxr-x--- 1 codespace codespace 4096 Jan  2 09:51 ..
-rw-rw-r-- 1 codespace codespace  0 Jan  2 09:53 entry-script.sh
-rw-rw-r-- 1 codespace codespace  0 Jan  2 09:53 locals.tf
-rw-rw-r-- 1 codespace codespace  0 Jan  2 09:53 main.tf
-rw-rw-r-- 1 codespace codespace  0 Jan  2 09:53 outputs.tf
-rw-rw-r-- 1 codespace codespace  0 Jan  2 09:53 terraform.tfvars
-rw-rw-r-- 1 codespace codespace  0 Jan  2 09:53 variables.tf
@toobashafique065 [ ] ~/Lab12 $
```

- task1_variables_tf.png

```
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "instance_type" {}
variable "public_key" {}
variable "private_key" {}
```

- task1_outputs_tf.png

```
output "aws_instance_public_ip" {
  value = aws_instance.myapp-server.public_ip
}
```

- task1_locals_tf.png

```
locals {
  my_ip = "${chomp(data.http.my_ip.response_body)}/32"
}

data "http" "my_ip" {
  url = "https://icanhazip.com"
}
```

- task1_terraform_tfvars.png

```
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/ssh/id_ed25519.pub"
private_key = "~/ssh/id_ed25519"
```

- task1_main_tf.png

```

tags = {
  Name = "${var.env_prefix}-default-sg"
}

resource "aws_key_pair" "ssh-key" {
  key_name     = "serverkey"
  public_key   = file(var.public_key)
}

resource "aws_instance" "myapp-server" {
  ami          = "ami-05524d6658fcf35b6"
  instance_type = var.instance_type
  subnet_id    = aws_subnet.myapp_subnet_1.id
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name      = aws_key_pair.ssh-key.key_name
  user_data     = file("./entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

- task1_entry_script.png

```

@toobashafique065 ~ /Lab12 $ nano entry-script.sh
@toobashafique065 ~ /Lab12 $ chmod +x entry-script.sh
@toobashafique065 ~ /Lab12 $

```

```

#!/bin/bash
set -e
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx

```

- task1_ssh_keygen.png

```

@toobashafique065 ~ /Lab12 $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:CRhH3WOpuAYwvBIddgziiqrDGG+1vdMhiHMouLL6UOM codespace@codespaces-fb3ddd
The key's randomart image is:
+--[ED25519 256]--+
|.O++O.O. . . |
|OO=..+ . = |
| O +. . . O . |
|+ . . .... |
|O.OO O .S |
|+O=.+ + . |
|**E+ + O . |
|Boo . o . |
|+=. .O |
+----[SHA256]-----+

```

- task1_terraform_init.png

```
@toobashafique065 ~ /Lab12 $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Finding latest version of hashicorp/http...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 ~ /Lab12 $
```

- task1_terraform_apply.png

```
@toobashafique065 ~ /Lab12 $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are i
following symbols:
+ create

Terraform will perform the following actions:

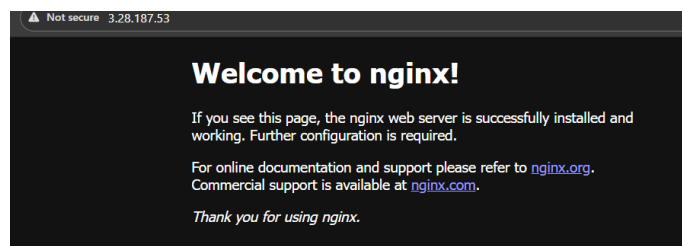
# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
+   arn                = (known after apply)
+   default_route_table_id = (known after apply)
+   id                 = (known after apply)
+   owner_id           = (known after apply)
+   region              = "me-central-1"
+   route               = [
+     + {
+       + cidr_block      = "0.0.0.0/0"
+       + gateway_id      = (known after apply)
+       # (10 unchanged attributes hidden)
+     },
+   ]
+   tags                = {
+     + "Name" = "dev-rt"
+   }
+   tags_all             = {
+     + "Name" = "dev-rt"
+   }
+   vpc_id              = (known after apply)
}

# aws_default_security_group.default_sg will be created
+ resource "aws_default_security_group" "default_sg" {
+   arn                = (known after apply)
```

- task1_terraform_output.png

```
@toobashafique065 ~ /Lab12 $ terraform output
aws_instance_public_ip = "3.28.187.53"
```

- task1_nginx_browser.png



- task1_terraform_destroy.png

```
aws_default_route_table.main_rt: Destroying... [id=rtb-067512c035dd3ae26]
aws_instance.myapp-server: Destroying... [id=i-050f5c0ab714a84c0]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-03e02f5c0be9a031e]
aws_instance.myapp-server: Still destroying... [id=i-050f5c0ab714a84c0, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-03e02f5c0be9a031e, 00m10s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-050f5c0ab714a84c0, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-03e02f5c0be9a031e, 00m20s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-050f5c0ab714a84c0, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-03e02f5c0be9a031e, 00m30s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-050f5c0ab714a84c0, 00m40s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-03e02f5c0be9a031e, 00m40s elapsed]
aws_instance.myapp-server: Destruction complete after 48s
aws_internet_gateway.myapp_igw: Destruction complete after 48s
aws_instance.myapp-server: Still destroying... [id=i-050f5c0ab714a84c0, 00m50s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-050f5c0ab714a84c0, 01m00s elapsed]
aws_instance.myapp-server: Destruction complete after 1m1s
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_default_security_group.default_sg: Destroying... [id=sg-06f8fa58c9583ea74]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-07b755749d39d55c4]
aws_default_security_group.default_sg: Destruction complete after 0s
aws_key_pair.ssh-key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-01552aca2b5627e21]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
root@kali:~/tf/095_Brew/1-ah12 $
```

Task 2 — Remote-exec Provisioner

Screenshots Required:

- task2_main_tf_remote_exec.png

```
resource "aws_instance" "myapp-server" {
  ami = "ami-05524d6658fcf35b6"
  instance_type = var.instance_type
  subnet_id = aws_subnet.myapp_subnet_1.id
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name = aws_key_pair.ssh_key.key_name

  connection {
    type     = "ssh"
    user     = "ec2-user"
    private_key = file(var.private_key)
    host     = self.public_ip
  }

  provisioner "remote-exec" {
    inline = [
      "sudo yum update -y",
      "sudo yum install -y nginx",
      "sudo systemctl start nginx",
      "sudo systemctl enable nginx"
    ]
  }
}
```

- task2_terraform_apply.png

```

aws_instance.myapp-server (remote-exec): Installing : generic- [====] 6/7
aws_instance.myapp-server (remote-exec): Installing : generic-logo 6/7
aws_instance.myapp-server (remote-exec): Installing : nginx [====] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx [====] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx [====] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx [====] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx-1:1.28 7/7
aws_instance.myapp-server (remote-exec): Running scriptlet: nginx-1:1.28 7/7
aws_instance.myapp-server: Still creating... [01m00s elapsed]
aws_instance.myapp-server (remote-exec): Verifying : generic-logo 1/7
aws_instance.myapp-server (remote-exec): Verifying : gperftools-1 2/7
aws_instance.myapp-server (remote-exec): Verifying : libunwind-1. 3/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-1:1.28 4/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-core-1 5/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-filesy 6/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-mimety 7/7

aws_instance.myapp-server (remote-exec): Installed:
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
aws_instance.myapp-server (remote-exec): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
aws_instance.myapp-server: Creation complete after 1m1s [id=i-03ad9102d8b622ed6]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

```

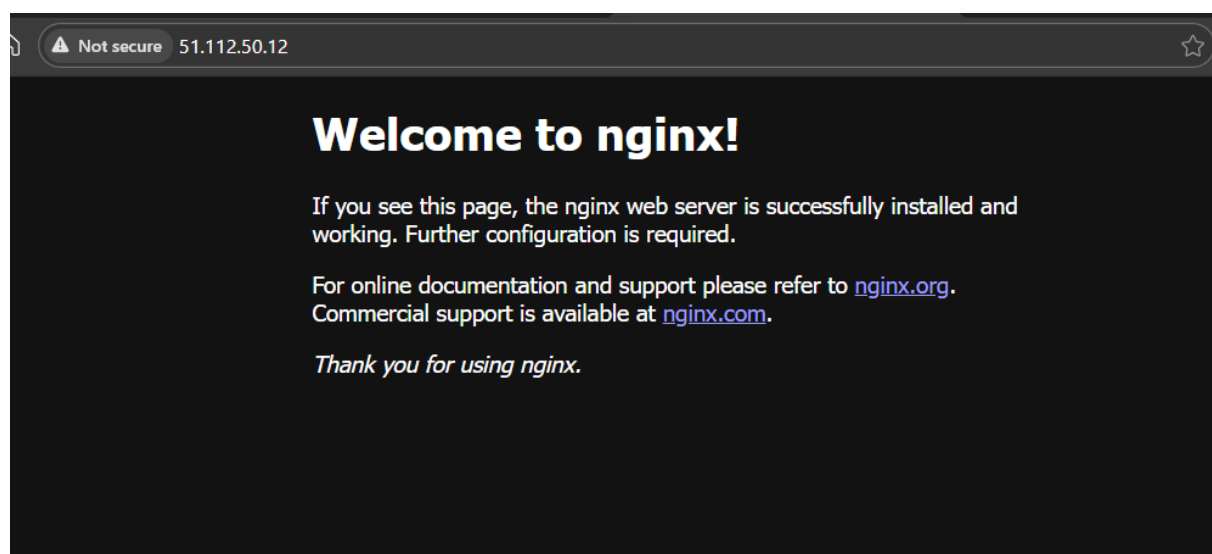
- task2_terraform_output.png

```

@toobashafique065 ~ /Lab12 $ terraform output
aws_instance_public_ip = "51.112.50.12"

```

- task2_nginx_browser.png



Task 3 — File & Local-exec Provisioners

Screenshots Required:

- task3_main_tf_all_provisioners.png

```
resource "aws_instance" "myapp-server" {
  ami           = "ami-05524d6658f35b6"
  instance_type = var.instance_type
  subnet_id     = aws_subnet.myapp_subnet_1.id
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name       = aws_key_pair.ssh-key.key_name

  connection {
    type     = "ssh"
    user     = "ec2-user"
    private_key = file(var.private_key)
    host     = self.public_ip
  }

  provisioner "file" {
    source     = "./entry-script.sh"
    destination = "/home/ec2-user/entry-script-on-ec2.sh"
  }

  provisioner "remote-exec" {
    inline = [
      "sudo chmod +x /home/ec2-user/entry-script-on-ec2.sh",
      "sudo /home/ec2-user/entry-script-on-ec2.sh"
    ]
  }

  provisioner "local-exec" {
    command = <<-EOF
    echo Instance ${self.id} with public IP ${self.public_ip} has been created
    EOF
  }

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}
```

- task3_terraform_apply.png

```
aws_instance.myapp-server (remote-exec): Installing : nginx [=== ] 5/7
aws_instance.myapp-server (remote-exec): Installing : nginx [==== ] 5/7
aws_instance.myapp-server (remote-exec): Installing : nginx [===== ] 5/7
aws_instance.myapp-server (remote-exec): Installing : nginx-core-1 5/7
aws_instance.myapp-server (remote-exec): Installing : gener [ ] 6/7
aws_instance.myapp-server (remote-exec): Installing : gener [==== ] 6/7
aws_instance.myapp-server (remote-exec): Installing : gener [===== ] 6/7
aws_instance.myapp-server (remote-exec): Installing : generic-logo 6/7
aws_instance.myapp-server (remote-exec): Installing : nginx [ ] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx [== ] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx [=== ] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx [===== ] 7/7
aws_instance.myapp-server (remote-exec): Installing : nginx-1:1.28 7/7
aws_instance.myapp-server (remote-exec): Running scriptlet: nginx-1:1.28 7/7
aws_instance.myapp-server (remote-exec): Verifying : generic-logo 1/7
aws_instance.myapp-server (remote-exec): Verifying : gperf-tools-1 2/7
aws_instance.myapp-server (remote-exec): Verifying : libunwind-1 3/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-1:1.28 4/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-core-1 5/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-filesys 6/7
aws_instance.myapp-server (remote-exec): Verifying : nginx-mimety 7/7

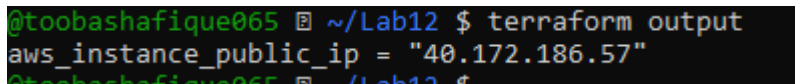
aws_instance.myapp-server (remote-exec): Installed:
aws_instance.myapp-server (remote-exec): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
aws_instance.myapp-server (remote-exec): gperf-tools-libs-2.9.1-1.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): libunwind-1.4.0-5.amzn2023.0.3.x86_64
aws_instance.myapp-server (remote-exec): nginx-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
aws_instance.myapp-server (remote-exec): nginx-filessystem-1:1.28.0-1.amzn2023.0.2.noarch
aws_instance.myapp-server (remote-exec): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

aws_instance.myapp-server (remote-exec): Complete!
aws_instance.myapp-server (remote-exec): Created symlink /etc/systemd/system/multi-user.target
aws_instance.myapp-server: Provisioning with 'local-exec'...
aws_instance.myapp-server (local-exec): Executing: ["/bin/sh" "-c" "echo Instance i-046e562139
aws_instance.myapp-server (local-exec): Instance i-046e5621398531ef5 with public IP 40.172.186
aws_instance.myapp-server: Creation complete after 1m29s [id=i-046e5621398531ef5]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

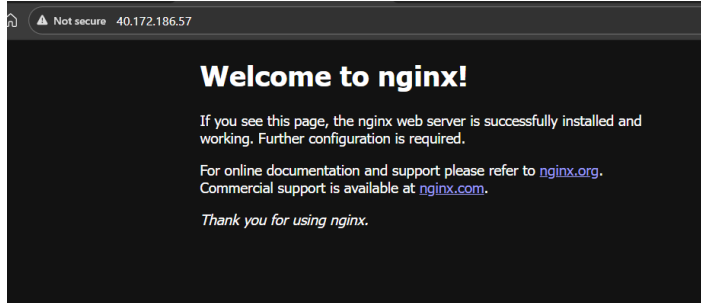
Outputs:
aws_instance_public_ip = "40.172.186.57"
```


- task3_terraform_output.png

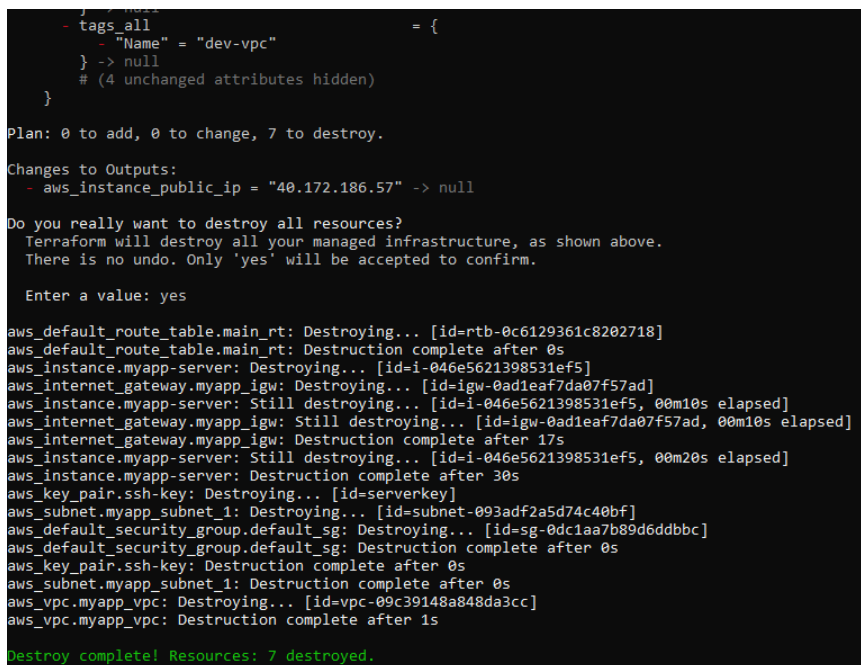


```
@toobashafique065 ~ /Lab12 $ terraform output
aws_instance_public_ip = "40.172.186.57"
```

- task3_nginx_browser.png



- task3_terraform_destroy.png



```
Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:
- aws_instance_public_ip = "40.172.186.57" -> null

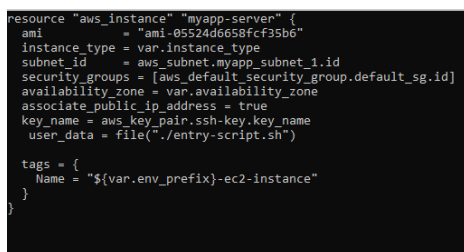
Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_default_route_table.main_rt: Destroying... [id=rtb-0c6129361c8202718]
aws_default_route_table.main_rt: Destruction complete after 0s
aws_instance.myapp-server: Destroying... [id=i-046e5621398531ef5]
aws_internet_gateway.myapp_igw: Destroying... [id=igw-0ad1eaf7da07f57ad]
aws_instance.myapp-server: Still destroying... [id=i-046e5621398531ef5, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0ad1eaf7da07f57ad, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 17s
aws_instance.myapp-server: Still destroying... [id=i-046e5621398531ef5, 00m20s elapsed]
aws_instance.myapp-server: Destruction complete after 30s
aws_key_pair.ssh-key: Destroying... [id=serverkey]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-093adf2a5d74c40bf]
aws_default_security_group.default_sg: Destroying... [id=sg-0dc1aa7b89d6ddbcb]
aws_default_security_group.default_sg: Destruction complete after 0s
aws_key_pair.ssh-key: Destruction complete after 0s
aws_subnet.myapp_subnet_1: Destruction complete after 0s
aws_vpc.myapp_vpc: Destroying... [id=vpc-09c39148a848da3cc]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
```

- task3_main_tf_restored.png



```
resource "aws_instance" "myapp-server" {
  ami           = "ami-05524d6658fcf35b6"
  instance_type = var.instance_type
  subnet_id     = aws_subnet.myapp_subnet_1.id
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name       = aws_key_pair.ssh-key.key_name
  user_data     = file("./entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}
```

Task 4 — Subnet Module

Screenshots Required:

- task4_module_structure.png


```
@toobashafique065 [~ /Lab12] $ mkdir -p modules/subnet
@toobashafique065 [~ /Lab12] $ touch modules/subnet/main.tf
@toobashafique065 [~ /Lab12] $ touch modules/subnet/variables.tf
@toobashafique065 [~ /Lab12] $ touch modules/subnet/outputs.tf
```

- task4_subnet_variables.png

```
GNU nano 7.2
variable "vpc_id" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
variable "default_route_table_id" {}
```

- task4_subnet_main.png

```
GNU nano 7.2
resource "aws_subnet" "myapp_subnet_1" {
  vpc_id = var.vpc_id
  cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  map_public_ip_on_launch = true
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_default_route_table" "main_rt" {
  default_route_table_id = var.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }
  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = var.vpc_id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}
```

- task4_subnet_outputs.png

```
GNU nano 7.2
output "subnet" {
  value = aws_subnet.myapp_subnet_1
}
```

- task4_main_tf_with_module.png

```
module "myapp-subnet" {
  source = "../modules/subnet"
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}
```

```

}
resource "aws_instance" "myapp-server" {
  ami           = "ami-05524d6658fc35b6" # Amazon Linux 2023 Kernel 6.1 AMI
  instance_type = var.instance_type
  subnet_id     = module.myapp-subnet.subnet.id # <-- updated
  security_groups = [aws_default_security_group.default_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name       = aws_key_pair.ssh-key.key_name

  user_data = file("./entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

```

- task4_terraform_init.png

```

@toobashafique065 @ ~/Lab12 $ terraform init
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 @ ~/Lab12 $

```

- task4_terraform_apply.png

```

Changes to Outputs:
  + aws_instance_public_ip = (known after apply)
aws_key_pair.ssh-key: Creating...
aws_vpc.myapp_vpc: Creating...
aws_key_pair.ssh-key: Creation complete after 0s [id=serverkey]
aws_vpc.myapp_vpc: Creation complete after 1s [id=vpc-01ca6c7af1c3a3a21]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
aws_default_security_group.default_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-04ba0277b64ca05e2]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-04ad56d7aed61dcf0]
aws_default_security_group.default_sg: Creation complete after 3s [id=sg-04471b558d3d86a5]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-0c6f2b66fea59104a]
aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still creating... [00m10s elapsed]
aws_instance.myapp-server: Creation complete after 13s [id=i-0e4beed7b957d06b4]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:
aws_instance_public_ip = "3.28.187.147"

```

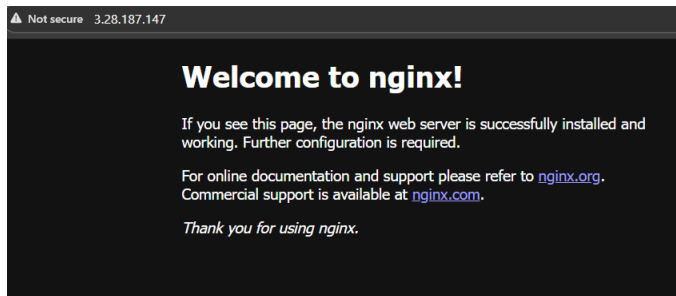
- task4_terraform_output.png

```

@toobashafique065 @ ~/Lab12 $ terraform output
aws_instance_public_ip = "3.28.187.147"

```

- task4_nginx_browser.png



Task 5 — Webserver Module

Screenshots Required:

- task5_webserver_module_structure.png

```
@toobashafique065 @ /workspaces/Lab12 (main) $ mkdir -p modules/webserver
@toobashafique065 @ /workspaces/Lab12 (main) $ touch modules/webserver/main.tf
@toobashafique065 @ /workspaces/Lab12 (main) $ touch modules/webserver/variables.tf
@toobashafique065 @ /workspaces/Lab12 (main) $ touch modules/webserver/outputs.tf
@toobashafique065 @ /workspaces/Lab12 (main) $
```

- task5_webserver_variables.png

```
variable "env_prefix" {}
variable "instance_type" {}
variable "availability_zone" {}
variable "public_key" {}
variable "my_ip" {}
variable "vpc_id" {}
variable "subnet_id" {}
variable "script_path" {}
variable "instance_suffix" {}
```

- task5_webserver_main.png

```
resource "aws_security_group" "web_sg" {
  vpc_id      = var.vpc_id
  name        = "${var.env_prefix}-web-sg-${var.instance_suffix}"
  description = "Security group for web server allowing HTTP, HTTPS and SSH"

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = [var.my_ip]
  }
  ingress {
    from_port = 443
    to_port   = 443
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
    prefix_list_ids = []
  }

  tags = {
    Name = "${var.env_prefix}-web-sg-${var.instance_suffix}"
  }
}

resource "aws_key_pair" "ssh-key" {
  key_name      = "${var.env_prefix}-serverkey-${var.instance_suffix}"
  public_key    = file(var.public_key)
}
```

- task5_webserver_outputs.png

```
GNU nano 7.2
output "aws_instance" {
  value = aws_instance.myapp-server
}
```

- task5_main_tf_webserver_module.png

```
# -----
# Webserver module
# -----
module "myapp-webserver" {
  source          = "../modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "../entry-script.sh"
  instance_suffix = ""
}
```

- task5_outputs_updated.png

```
output "webserver_public_ip" {
  value = module.myapp-webserver.aws_instance.public_ip
}
```

- task5_terraform_init.png

```
@toobashafique065 ~ ~/Lab12 $ terraform init
Initializing the backend...
Initializing modules...
- myapp-webserver in modules/webserver
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/http v3.5.0
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@toobashafique065 ~ ~/Lab12 $
```

- task5_terraform_apply.png

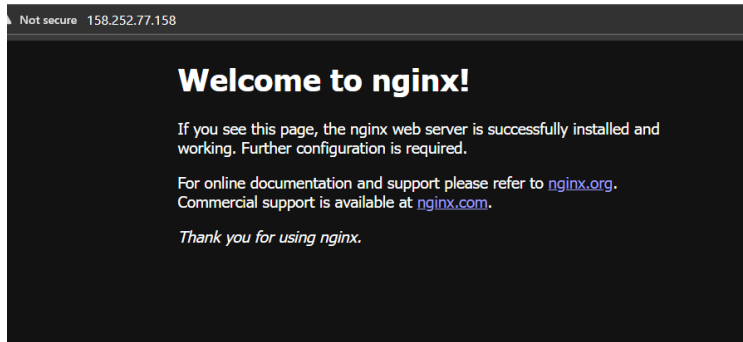
```
aws_instance.myapp-server: Destroying... [id=i-09cd4fcceb2c0673]
module.myapp-webserver.aws_key_pair.ssh-key: Creating...
module.myapp-webserver.aws_security_group.web_sg: Creating...
module.myapp-webserver.aws_key_pair.ssh-key: Creation complete after 1s [id=dev-serverkey-0]
module.myapp-webserver.aws_security_group.web_sg: Creation complete after 3s [id=sg-014ce72b10e0e0e9]
module.myapp-webserver.aws_instance.myapp-server: Creating...
aws_instance.myapp-server: Still destroying... [id=i-09cd4fcceb2c0673, 00m10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Creation complete after 11s [id=i-027e068381140c1bd]
aws_instance.myapp-server: Still destroying... [id=i-09cd4fcceb2c0673, 00m20s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-09cd4fcceb2c0673, 00m30s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-09cd4fcceb2c0673, 00m40s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-09cd4fcceb2c0673, 00m50s elapsed]
aws_instance.myapp-server: Still destroying... [id=i-09cd4fcceb2c0673, 01m00s elapsed]
aws_instance.myapp-server: Destruction complete after 1m1s
aws_key_pair.ssh-key: Destroying... [id=dev-serverkey]
aws_default_security_group.default_sg: Destroying... [id=sg-04471b558d38d86a5]
aws_default_security_group.default_sg: Destruction complete after 0s
aws_key_pair.ssh-key: Destruction complete after 0s

Apply complete! Resources: 3 added, 0 changed, 3 destroyed.
```

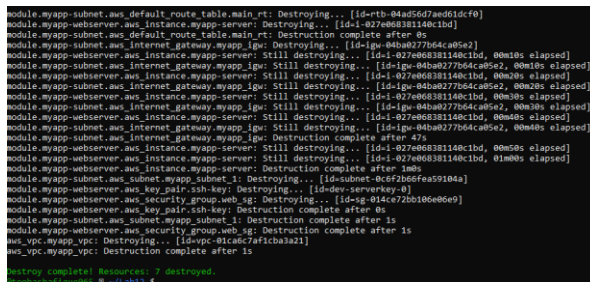
- task5_terraform_output.png

```
@toobashafique065 ~ ~/Lab12 $ terraform output
webserver_public_ip = "158.252.77.158"
```

- task5_nginx_browser.png



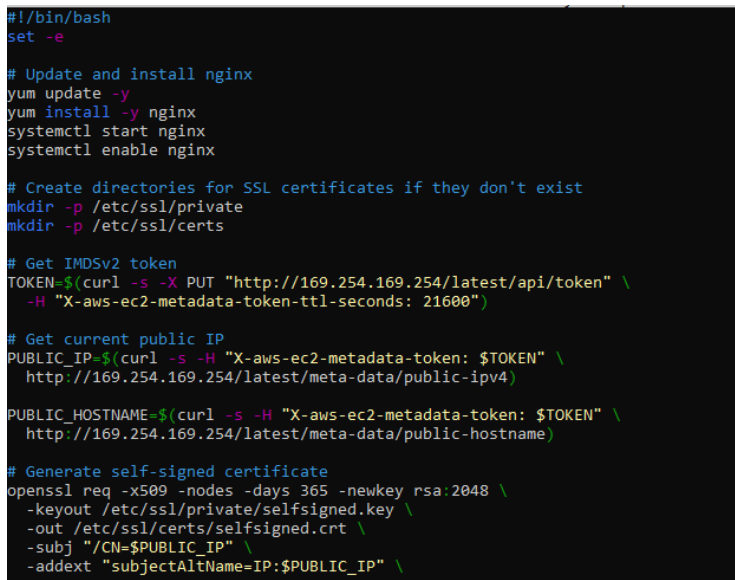
- task5_terraform_destroy.png



Task 6 — HTTPS with Self-signed Certificates

Screenshots Required:

- task6_entry_script_https.png



- task6_terraform_apply.png

```

    + "Name" = "dev-default-sg"
  }
  + vpc_id      = (known after apply)
}

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  - webserver_public_ip = (known after apply)
module.myapp-webserver.aws_key_pair.ssh-key: Creating...
aws_vpc.myapp_vpc: Creating...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-webserver.aws_security_group.web_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 0s [id=igw-0f40825861c97095b]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-058d351275f1fe70a]
module.myapp-subnet.aws_security_group.web_sg: Creation complete after 3s [id=sg-07b860a2e56a0725b]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-08b93f4244c3da952]
module.myapp-webserver.aws_instance.myapp-server: Creating...
module.myapp-webserver.aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver.aws_instance.myapp-server: Creation complete after 13s [id=i-0f3ae4faf591c0d53]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:
webserver_public_ip = "3.28.131.42"

```

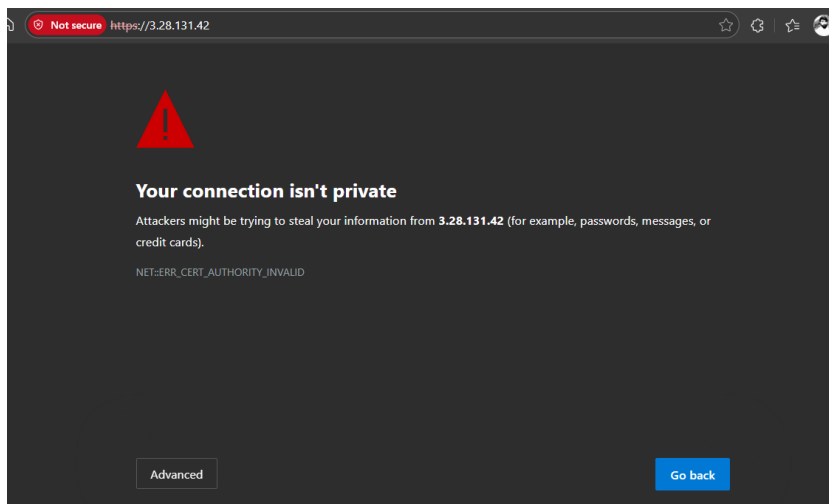
- task6_terraform_output.png

```

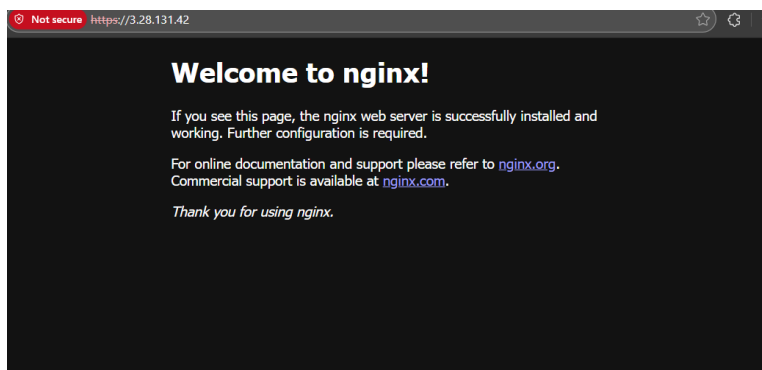
@toobashafique065 ~ /Lab12 $ terraform output
webserver_public_ip = "3.28.131.42"

```

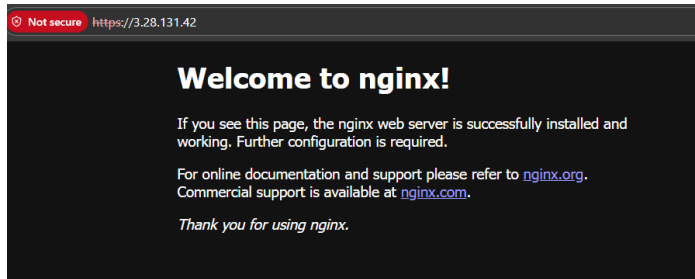
- task6_browser_security_warning.png



- task6_nginx_https_browser.png



- task6_http_redirect.png



Task 7 — Nginx Reverse Proxy

Screenshots Required:

- task7_apache_script.png

```
GNU nano 2.9.2 apache.sh
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "<h1>Welcome to My Web Server</h1>" > /var/www/html/index.html
hostnamectl set-hostname myapp-webserver

TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")

echo "<h2>Hostname: $(hostname)</h2>" >> /var/www/html/index.html
echo "<h2>Private IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/local-ipv4)</h2>" >> /var/www/html/index.html
echo "<h2>Public IP: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-ipv4)</h2>" >> /var/www/html/index.html
echo "<h2>Public DNS: $(curl -s -H "X-aws-ec2-metadata-token: $TOKEN" http://169.254.169.254/latest/meta-data/public-hostname)</h2>" >> /var/www/html/index.html
echo "<h2>Deployed via Terraform</h2>" >> /var/www/html/index.html
```

- task7_main_tf_web1.png

```
module "myapp-web-1" {
  source          = "../modules/webserver"
  env_prefix      = var.env_prefix
  instance_type   = var.instance_type
  availability_zone = var.availability_zone
  public_key      = var.public_key
  my_ip           = local.my_ip
  vpc_id          = aws_vpc.myapp_vpc.id
  subnet_id       = module.myapp-subnet.subnet.id
  script_path     = "../apache.sh"
  instance_suffix = "1"
}
```

- task7_outputs_web1.png

```
output "aws_web1_public_ip" {
  value = module.myapp-web-1.aws_instance.public_ip
}
```

- task7_terraform_apply.png

```
> owner_id           = (known after apply)
> region             = "me-central-1"
> revoke_rules_on_delete = false
> tags               = {
  + "Name" = "dev-default-sg"
}
> tags_all           = {
  + "Name" = "dev-default-sg"
}
> vpc_id             = "vpc-0e7ce985e88420e5f"
}

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + aws_web1_public_ip = (known after apply)
module.myapp-web-1.aws_key_pair.ssh-key: Creating...
module.myapp-web-1.aws_security_group.web_sg: Creating...
module.myapp-web-1.aws_key_pair.ssh-key: Creation complete after 0s [id-dev-serverkey-1]
module.myapp-web-1.aws_security_group.web_sg: Creation complete after 2s [id-sg-08559831aa6b13d29]
module.myapp-web-1.aws_instance.myapp-server: Creating...
module.myapp-web-1.aws_instance.myapp-server: Still creating... [00m18s elapsed]
module.myapp-web-1.aws_instance.myapp-server: Creation complete after 12s [id-i-07f583a8e71f2585c]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```


- task7_terraform_output.png 3.28.131.42

```
@toobashafique065 ~ /Lab12 $ terraform output
aws_web1_public_ip = "51.112.51.132"
webserver_public_ip = "3.28.131.42"
@toobashafique065 ~ /Lab12 $
```

- task7_ssh_webserver.png

```
@toobashafique065 ~ /Lab12 $ ssh ec2-user@3.28.131.42
The authenticity of host '3.28.131.42 (3.28.131.42)' can't be established.
ED25519 key fingerprint is SHA256:s6pvIIR+P0LDqsTLnmmdVcwnPQIMX+bpCIfIvrVtjo8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.28.131.42' (ED25519) to the list of known hosts.

      ##
     ###      Amazon Linux 2023
    #####
   #####\
  ##### \###
  #####  \##/
  #####   V##
  #####   /
  #####  /
  ##### /m/

[ec2-user@ip-10-0-10-11 ~]$
```

- task7_nginx_conf_reverse_proxy.png

```
ssl_certificate_key /etc/ssl/private/selfsigned.key;

location / {
    proxy_pass http://51.112.51.132:80;
    #index index.html;
}
}
```

- task7_nginx_restart.png

```
[ec2-user@ip-10-0-10-11 ~]$ sudo systemctl restart nginx
```

- task7_error_log.png

```
[ec2-user@ip-10-0-10-11 ~]$ cat /var/log/nginx/error.log
2026/01/02 14:55:57 [notice] 2944#2944: using the "epoll" event method
2026/01/02 14:55:57 [notice] 2944#2944: nginx/1.28.0
2026/01/02 14:55:57 [notice] 2944#2944: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/02 14:55:57 [notice] 2944#2944: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/02 14:55:57 [notice] 2993#2993: start worker processes
2026/01/02 14:55:57 [notice] 2993#2993: start worker process 2995
2026/01/02 14:55:57 [notice] 2993#2993: start worker process 2996
2026/01/02 14:55:57 [notice] 2993#2993: signal 3 (SIGQUIT) received from 1, shutting down
2026/01/02 14:55:57 [notice] 2995#2995: gracefully shutting down
2026/01/02 14:55:57 [notice] 2995#2995: exiting
2026/01/02 14:55:57 [notice] 2995#2995: exit
2026/01/02 14:55:57 [notice] 2996#2996: gracefully shutting down
2026/01/02 14:55:57 [notice] 2996#2996: exiting
2026/01/02 14:55:57 [notice] 2996#2996: exit
2026/01/02 14:55:57 [notice] 2993#2993: signal 17 (SIGCHLD) received from 2995
2026/01/02 14:55:57 [notice] 2993#2993: worker process 2995 exited with code 0
2026/01/02 14:55:57 [notice] 2993#2993: signal 29 (SIGIO) received
2026/01/02 14:55:57 [notice] 2993#2993: signal 17 (SIGCHLD) received from 2996
2026/01/02 14:55:57 [notice] 2993#2993: worker process 2996 exited with code 0
2026/01/02 14:55:57 [notice] 2993#2993: exit
2026/01/02 14:55:57 [notice] 3759#3759: using the "epoll" event method
2026/01/02 14:55:57 [notice] 3759#3759: nginx/1.28.0
2026/01/02 14:55:57 [notice] 3759#3759: OS: Linux 6.1.158-180.294.amzn2023.x86_64
2026/01/02 14:55:57 [notice] 3759#3759: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/01/02 14:55:57 [notice] 3793#3793: start worker processes
2026/01/02 14:55:57 [notice] 3793#3793: start worker process 3797
2026/01/02 14:55:57 [notice] 3793#3793: start worker process 3798
2026/01/02 14:57:36 [error] 3798#3798: *5 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 175.107.227.5, server: 3.28.131.42, request: "GET /favicon.ico HTTP/1.1", host: "3.28
```

- task7_access_log.png

```
[ec2-user@ip-10-0-10-11 ~]$ cat /var/log/nginx/access.log
175.107.227.5 - - [02/Jan/2026:14:57:36 +0000] "GET / HTTP/1.1"200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0" "-"
175.107.227.5 - - [02/Jan/2026:14:57:36 +0000] "GET /favicon.ico HTTP/1.1"404 555 "https://3.28.131.42/"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0" "-"
175.107.227.5 - - [02/Jan/2026:14:58:52 +0000] "GET / HTTP/1.1"200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0" "-"
175.107.227.5 - - [02/Jan/2026:14:59:21 +0000] "GET / HTTP/1.1"200 615 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0" "-"
91.224.92.121 - - [02/Jan/2026:15:06:01 +0000] "OPTIONS / HTTP/1.1"301 169 "-"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/114.0.5735.199 Safari/537.36" "-"
204.76.203.20 - - [02/Jan/2026:15:09:02 +0000] "GET /index.asp HTTP/1.1"301 169 "-"Hello World/1.0" "-"
89.42.231.179 - - [02/Jan/2026:15:12:00 +0000] "GET / HTTP/1.1"301 169 "-"Mozilla/5.0" "-"
176.62.183.186 - - [02/Jan/2026:15:12:55 +0000] "POST /cgi-bin/.%2e/.%2e/.%2e/.%2e/.%2e/.%2e/bin/ HTTP/1.1"400 157 "-"-" "-"
176.62.183.186 - - [02/Jan/2026:15:12:56 +0000] "POST /cgi-bin/%32%65%32%65/%32%65%32%65/%32%65%32%65/bin/sh HTTP/1.1"400 157 "-"-" "-"
176.62.183.186 - - [02/Jan/2026:15:12:56 +0000] "POST /hello.world?%Add+allow_url_include%3d1+%Add+auto_prepend_file%hp://input HTTP/1.1"301 169 "-"libredtail-http" "-"
176.62.183.186 - - [02/Jan/2026:15:12:57 +0000] "POST /hello.world?%Add+allow_url_include%3d1+%Add+auto_prepend_file%hp://input HTTP/1.1"404 153 "-"libredtail-http" "-"
176.62.183.186 - - [02/Jan/2026:15:12:57 +0000] "POST /?%Add+allow_url_include%3d1+%Add+auto_prepend_file%3dphp://inp HTTP/1.1"405 157 "-"libredtail-http" "-"
176.62.183.186 - - [02/Jan/2026:15:12:57 +0000] "GET /vendor/phpunit/phpunit/src/Util/PHP/eval-stdin.php HTTP/1.1"404 3 "-"libredtail-http" "-"
[ec2-user@ip-10-0-10-11 ~]$
```

- task7_mime_types.png

```
[ec2-user@ip-10-0-10-11 ~]$ cat /etc/nginx/mime.types
types {
application/A2L                                a2l;
application/AML                                aml;
application/andrew-inset                       ez;
application/ATF                                atf;
application/ATFX                               atfx;
application/ATXML                             atxml;
application/atom+xml                          atom;
application/atomcat+xml                       atomcat;
application/atomdeleted+xml                   atomdeleted;
application/atomsvc+xml                       atomsvc;
application/atssc-dwd+xml                     dwd;
application/atssc-held+xml                     held;
application/atssc-rsat+xml                     rsat;
application/auth-policy+xml                   apxml;
application/bacnet-xdd+zip                     xdd;
application/calendar+xml                      xcs;
application/cbor                               cbor;
application/cccx                                c3ex;
application/ccmp+xml                          ccmp;
application/ccxml+xml                         ccxml;
application/CDFX+XML                          cdfx;
application/cdmi-capability                   cdmia;
application/cdmi-container                     cdmic;
application/cdmi-domain                       cdmid;
application/cdmi-object                       cdmio;
application/cdmi-queue                         cdmiq;
application/CEA                                cea;
application/cellml+xml                         cellml cml;
application/clue_info+xml                     clue;
application/cms                                cmsc;
application/cpl+xml                           cpl;
```

- task7_ssl_cert.png

```
[ec2-user@ip-10-0-10-11 ~]$ cat /etc/ssl/certs/selfsigned.crt
-----BEGIN CERTIFICATE-----
MIID0zCCAiOgAwIBAgIU58s1yv5e1GEFp1QSTBq6tv8iJ+0wDQYJKoZIhvcNAQEL
BQAwFjEUMBIGA1UEAwMLMy4yOC4xMzEuNDIwHhcNMjYwMTAyMTQ1NTU3WhcNMjcw
MTAyMTQ1NTU3WjAwMRQwEgYDVQQDDAszLjI4LjEzMS40MjCCAS1wDQYJKoZIhvcN
AQEBBQADggEPADCCAQoCggEBAMUvJoiEsbAkVprTRmxcmhPpABGcpJJBPxjzAws
0H/E0kOo4CrzBLzId8gsGOf+kr07SG6YsWoQQD79SVfjpJngzLw0urM/6iZSKh61
8ehCeip41Bk1JhYA88j3P6vCAd5Rkvr+o2G8YS1Z3ph+hTmr8Fhtu2uM+4hfB8bdT
yA0UKU/sVN9oF9sJfF6WnmKmxFSj+0eHFxmN0dM2n1rzIHxckZGBGHIe4/4ZCFvqJ
mLyNJ3Hb9E6KE16Axj5COhh6bp1kralf1QO2SDbNH9x5f+kIgsRepLHXJrx7Kb1Eu
0Wz3o8M53K4Nwt1COTZgIFFg1oSr4qPXqgjAyccCCp31PMkCAwEAAaOBgDB+MB0G
A1UdDgQWBBrTbrQopuSjE4EcMyIu2L4Gx0gVGXTAFBgNVHSMGDAWgBTbrQopuSjE
4EcMyIu2L4Gx0gVGXTAPBgNVHREEDAGHwQDHIMqMAkGA1UdEwQCMAAwCwYDVDR0P
BAQDAgWgMBGA1UdJQMMAAGCCsGAQUFBwMBMA0GCsQGS1b3DQEBcwUAA4IBAQCf
1hiqd9CP05kdV9VF1E4eE3/qh5yQ6zo1a45bKQoeee490SBeI2TqU7qjKTaz786U
Z39qLBzK4TRsRAJKh6C/9d2RgdmHemE1eajjgOV1cwwuudN3fM0WT0Hmf9AQ4mTT
0tChfRjUekum3mfCi+Iqel26mxPGqVsx3uV50X8Fws134Eo2QBfjQFB1jSWibmD
EaeUEG08sOLv1ytlV1Q1c02akp0J2KTH2f21Sekka/HB2/CkPUmwMAhScn7qmhlg
ybyj1GIf+yCrFJAyKnNhWtV51LTX1W12T4sNEp00YFCEx6kzcVD2M91WXTokaAN
1LjhRhb0rx0yW7rL8w6J
-----END CERTIFICATE-----
[ec2-user@ip-10-0-10-11 ~]$
```

- task7_ssl_key.png

```
[ec2-user@ip-10-0-10-11 ~]$ sudo cat /etc/ssl/private/selfsigned.key
-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBKgwggSkAgEAAoIBAQQDfLLY6NRLGwJFa
a00zsXJoT6QARnKSSQacY8wMLNB/xNJDQ0Aq8wS8yHfILBjn/pKzu0humLFqEEA+
/U1X46S4MYBNLqzP+omUpIetfHoQnoqeJQZJ5YVAPPI9z+rwgHeUZL/qNhhvGETw
d6Yf0U5q/BYbbrnjPuIXwFG3U8gNFCrv7FTfaBfBIxe1p51psRUo/tHhxV5jdHTN
p5a8yB13JGRgYByHuP+GQhb6iZi8jSSR2/R0ihJegMY+QjoYem6ZZK2n9UDTKg2z
R/ceX/pCILEXqSx1ya8eym9RLtFs96PD0dyuDcLdQjk2YCBRYJaEq+Kj16qowMnH
Agqd9TzJAgMBAAECggEAAAXfe3IAVq7IzYuD5qf9QQsaVCgt8PuqcSF3UPyL2Sn1E
s4xGxwAANYeonKz3YoLaxMvYNjqc15VXZSctsyHWz+6E18IswbjwHwJv/ZqN0pWE
I1z5FX1xUyieqBhvOPAw417oIU84jVs49GPfsMyasOE/q0Sibj7DQD0Jerv8+Vvx
ujw8XwSwsHBPzU0fsKK5BIRfyr/iJoTEvcz6F4x7U/qLG0GTErt7hBkkVY1oNhokR
ESNKxM/fdcNRHNzW29+IGLUABmivkKDBKTZeQf/b3+TEZ0Bz9+2ox76cWc6jEePt
KKQqWzyQ96TrYHCu8JqJ+e5N86mVN1TQ9PJqSavjoQKBgQD0ntbtjohGUXqAKum8
7DnMAa1Pp01DqrImzA4xBKjPOBYJ4Hi8m5WzZEEJmSLzHDu0N5uOUMDGTB1JEEzy
eGwaS1+fM1jX8RGietTJU1pP14jvF+kEzcwSot+DuJHTPoDnqmg9bGa4XD15d7K
jUFu1YZG8T3hHD6p4L5UtnIf8wKBgQDOWoukjsDMbD/8mpyUFFTDNswZnXbExwUd
hMLFYbZTxXjruRcSePVJ1kZ4sW8a02rT6tmgsNwJw1C8B6mqsbZrQ/PiEcBARj0a
AsoIgFsGphht3wjkg0rG7QY81KQ8dHohF0Z2ybZHMUDGzBDVF80StD7Zndx5V0J/
f8Y12+DbUwKBgQDvAqv53MqAMjyrHtJdn7CZ/RsuARQMdSzf1Hx8exA3n3+I2/
qz0W0s3RnQFD0JkBUzV4Z8qT27DB8m5JwanJHgeU0gvC6jXsXzVvGm2bKXswMeq
JuRMMVamSqcC5KF8G/CdxrJMjG4ctQdkXXRL0mnCsz/EzbYkgeU2HDW5JQKBG60D
TI5Mk/HqpgsQqvVXoEfY4Z77nsjp0vv7LOIVcrrisbTL/RhXKk4tkuEj62wikGcP
xJStmjDV13GKI5n4RER615a8hfHuqZaeFzxYiUrDfDMS0dfrNtWUFsks1fJa2SI
SULP4SnkLuJW3/WTL1CMRD+2s81tXVK57754QFvdAoGBANc6D3Fwz59IwXmuxF0Q
F0ThqGK6Adc6c6IFFnQyJNiJTTvy1NCm599rqP/IsWr2C1G2sVn0oYY/Dkf1/Sfk
y2N1FRjkw1oZkVIRCXjwDyBXXiZRHbCLSZQ+U/uDRakrxcD5eunSj9+wkYwTUTR
rRPIa0BZ+jtpQw0yxPNw5Uep
-----END PRIVATE KEY-----
```

- task7_reverse_proxy_browser.png



Task 8 — Nginx Load Balancer

Screenshots Required:

- task8_main_tf_web2.png

```

module "myapp-web-2" {
  source           = "../modules/webserver"
  env_prefix       = var.env_prefix
  instance_type    = var.instance_type
  availability_zone = var.availability_zone
  public_key       = var.public_key
  my_ip            = local.my_ip
  vpc_id           = aws_vpc.myapp_vpc.id
  subnet_id        = module.myapp-subnet.subnet.id
  script_path      = "../apache.sh"
  instance_suffix  = "2"
}

```

- task8_outputs_web2.png

```

output "aws_web-2_public_ip" {
  value = module.myapp-web-2.aws_instance.public_ip
}

```

- task8_terraform_apply.png

```

    ]
+ name           = "dev-web-sg-2"
+ name_prefix    = (known after apply)
+ owner_id       = (known after apply)
+ region         = "me-central-1"
+ revoke_rules_on_delete = false
+ tags           = {
+   + "Name" = "dev-default-sg"
+ }
+ tags_all       = {
+   + "Name" = "dev-default-sg"
+ }
+ vpc_id         = "vpc-0e7ce985e80420e5f"
}

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ aws_web-2_public_ip = (known after apply)
module.myapp-web-2.aws_key_pair.ssh-key: Creating...
module.myapp-web-2.aws_security_group.web_sg: Creating...
module.myapp-web-2.aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-2]
module.myapp-web-2.aws_security_group.web_sg: Creation complete after 3s [id=sg-02a591fb3948b0372]
module.myapp-web-2.aws_instance.myapp-server: Creating...
module.myapp-web-2.aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-web-2.aws_instance.myapp-server: Creation complete after 12s [id=i-0394c089bc31e1f53]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:
aws_web-2_public_ip = "3.28.191.12"
aws_web1_public_ip = "51.112.51.132"
webserver_public_ip = "3.28.131.42"

```

- task8_terraform_output.png

aws_web-2_public_ip = "3.28.191.12"

aws_web1_public_ip = "51.112.51.132"

webserver_public_ip = "3.28.131.42"

```

@toobashafique065 ~ /Lab12 $ terraform output
aws_web-2_public_ip = "3.28.191.12"
aws_web1_public_ip = "51.112.51.132"
webserver_public_ip = "3.28.131.42"

```

- task8_nginx_conf_load_balancer.png

```

upstream backend_servers {
    server 51.112.51.132:80;
    server 3.28.191.12:80;
}

server {
    listen 443 ssl;
    server_name 3.28.131.42;
    ssl_certificate /etc/ssl/certs/selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/selfsigned.key;

    location / {
        proxy_pass http://backend_servers;
        #index index.html;
    }
}

```

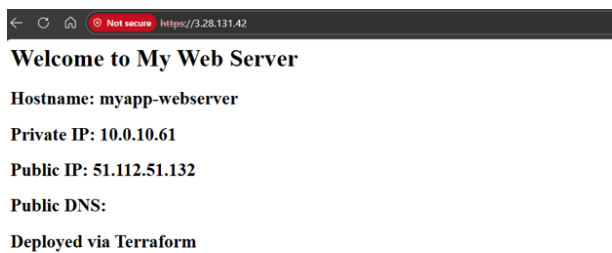
- task8_nginx_restart.png

```

[ec2-user@ip-10-0-10-11 ~]$ sudo systemctl restart nginx
[ec2-user@ip-10-0-10-11 ~]$

```

- task8_load_balancer_web1.png



← 🔍 🏠 Not secure https://3.28.131.42

Welcome to My Web Server

Hostname: myapp-webserver

Private IP: 10.0.10.61

Public IP: 51.112.51.132

Public DNS:

Deployed via Terraform

- task8_load_balancer_web2.png



← 🔍 🏠 Not secure https://3.28.131.42

Welcome to My Web Server

Hostname: myapp-webserver

Private IP: 10.0.10.24

Public IP: 3.28.191.12

Public DNS:

Deployed via Terraform

Task 9 — High Availability (Backup Servers)

Screenshots Required:

- task9_nginx_conf_ha_web1_primary.png

```

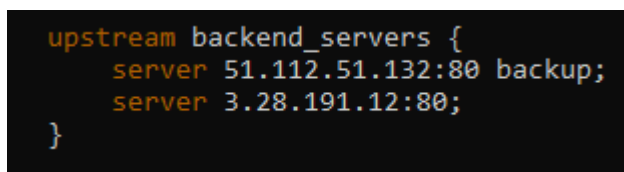
upstream backend_servers {
    server 51.112.51.132:80;
    server 3.28.191.12:80 backup;
}

```

- task9_ha_web1_only.png



- task9_nginx_conf_ha_web2_primary.png



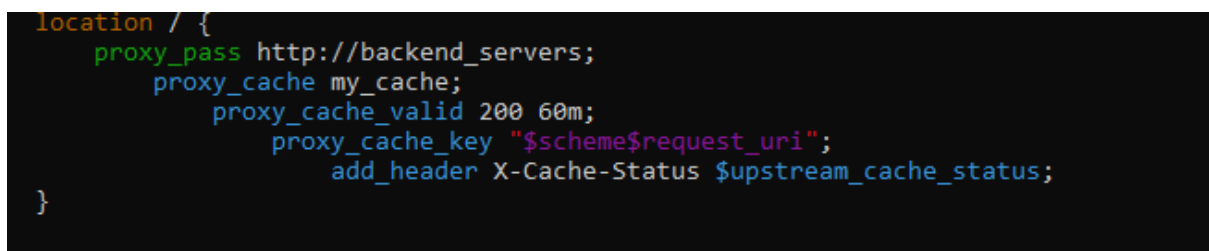
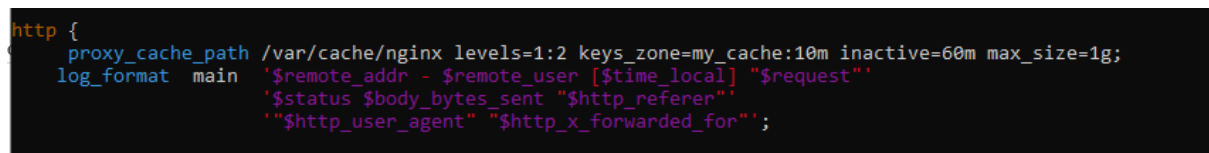
- task9_ha_web2_only.png



Task 10 — Nginx Caching

Screenshots Required:

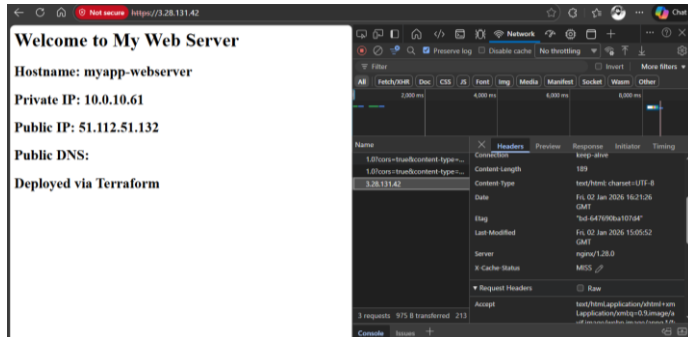
- task10_nginx_conf_cache.png



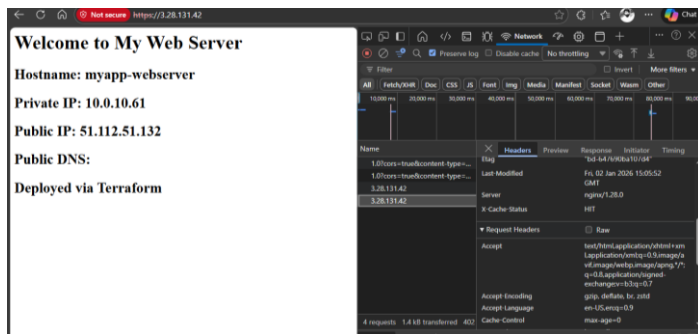
- task10_nginx_restart.png

```
[ec2-user@ip-10-0-10-11 ~]$ [ec2-user@ip-10-0-10-11 ~]$ sudo systemctl restart nginx
```

- task10_cache_miss.png



- task10_cache_hit.png



- task10_cache_directory.png

```
[ec2-user@ip-10-0-10-11 ~]$ sudo ls -la /var/cache/nginx
total 0
drwx-----. 3 nginx nginx 15 Jan  2 16:21 .
drwxr-xr-x. 9 root  root 101 Jan  2 16:13 ..
drwx-----. 3 nginx nginx 16 Jan  2 16:21 4
```

Cleanup

Screenshots Required:

- cleanup_destroy_prompt.png


```

@toobashafique065 ~ /Lab12 $ terraform destroy
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
module.myapp-web-1.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-1]
module.myapp-webserver.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
module.myapp-web-2.aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-2]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0e7ce985e80420e5f]
module.myapp-web-2.aws_security_group.web_sg: Refreshing state... [id=sg-02a591fb3948b0372]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-08b93f4244c3da952]
module.myapp-web-1.aws_security_group.web_sg: Refreshing state... [id=sg-08559831aa6b13d29]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0f40825861c97095b]
module.myapp-webserver.aws_security_group.web_sg: Refreshing state... [id=sg-07b8d0a2e56a0725b]
module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-058d351275f1fe70a]
module.myapp-web-2.aws_instance.myapp-server: Refreshing state... [id=i-0304c080bc31e1f53]
module.myapp-web-1.aws_instance.myapp-server: Refreshing state... [id=i-07f583a8e71f2585c]
module.myapp-webserver.aws_instance.myapp-server: Refreshing state... [id=i-0f3ae4faf591c0d53]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with
the following symbols:
- destroy

Terraform will perform the following actions:

# aws_vpc.myapp_vpc will be destroyed
- resource "aws_vpc" "myapp_vpc" {
  - arn = "arn:aws:ec2:me-central-1:535942231766:vpc/vpc-0e7ce985e80420e5f" ->
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-0328911c6f03a758e" -> null
  - default_route_table_id = "rtb-058d351275f1fe70a" -> null
  - default_security_group_id = "sg-00fa69b05103c0f52" -> null
  - dhcp_options_id = "dopt-02c0a1e8f3dd284d6" -> null
  - enable_dns_hostnames = false -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
}

```

- cleanup_destroy_complete.png

```

module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying...
module.myapp-web-2.aws_key_pair.ssh-key: Destroying...
module.myapp-web-2.aws_security_group.web_sg: Destroying...
module.myapp-web-2.aws_key_pair.ssh-key: Destroying...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying...
module.myapp-web-2.aws_security_group.web_sg: Destroying...
aws_vpc.myapp_vpc: Destroying... [id=vpc-0e7ce985e80420e5f]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 13 destroyed.
@toobashafique065 ~ /Lab12 $

```

- cleanup_state_empty.png

```

@toobashafique065 ~ /Lab12 $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 31,
  "lineage": "6d6b12d8-f5fb-1787-f4c5-fc11072d5066",
  "outputs": {},
  "resources": [],
  "check_results": null
}

```

- cleanup_final_files.png

```

@toobashafique065 ~ /Lab12 $ ls -la
total 100
drwxrwxr-x 4 codespace codespace 4096 Jan  2 16:30 .
drwxr-x--- 1 codespace codespace 4096 Jan  2 10:23 ..
drwxr-xr-x 4 codespace codespace 4096 Jan  2 11:27 .terraform
-rw-r--r-- 1 codespace codespace 2422 Jan  2 10:36 .terraform.lock.hcl
-rwxrwxr-x 1 codespace codespace  930 Jan  2 15:02 apache.sh
-rwxrwxr-x 1 codespace codespace 2442 Jan  2 14:54 entry-script.sh
-rw-rw-r-- 1 codespace codespace  125 Jan  2 14:54 locals.tf
-rw-rw-r-- 1 codespace codespace 2150 Jan  2 15:35 main.tf
drwxrwxr-x 4 codespace codespace 4096 Jan  2 14:32 modules
-rw-rw-r-- 1 codespace codespace  259 Jan  2 15:36 outputs.tf
-rw-rw-r-- 1 codespace codespace  182 Jan  2 16:30 terraform.tfstate
-rw-rw-r-- 1 codespace codespace 42635 Jan  2 16:29 terraform.tfstate.backup
-rw-rw-r-- 1 codespace codespace  220 Jan  2 10:43 terraform.tfvars
-rw-rw-r-- 1 codespace codespace  198 Jan  2 10:25 variables.tf

```